ALEX-TRONIX

A NEW WORLD OF CONTROLS

DOE/EE/15698-1

June 12, 1999

Mr Ramesh Jain
Invention and Innovation, EE-24
U.S. Dept of Energy
1000 Independence Ave, SW
Washington, D.C. 20585

Dear Mr Jain:

In accordance with your request and the agreed upon extended completion date of June 15, here is the final report of grant #DE-FG01-97EE15698. The other requested information is being forwarded to the appropriate offices along with the final request for funding in order to close out this grant.

With your indulgence, I would like to review the history of this project, showing the stages of development and how the DOE grant assisted in this evolution.

The product (the SWELL) originally started as an encapsulated circuit attached to a 24 VAC solenoid for irrigation systems to reduce the current draw of the solenoid, thereby saving energy and copper wiring cost. Patent #4,716,490 from 1987 covered this design. The energy saved was estimated at about 70% versus a standard AC solenoid. The wiring cost depended on the distances involved, but could be as much as 60%.

Unknown to us at the time, an Australian design similar to the SWELL had received a U.S. patent a few months prior to mine. This fact was brought to our attention shortly thereafter and prevented us from marketing our product.

In 1994, the SWELL was redesigned using a latching solenoid (one that requires one DC pulse to open the valve and one to close it with no power required to hold the valve open). The energy saving using the latching solenoid approached 90% and because it used a latching solenoid with AC voltage, it bypassed the Australian patent. Patent #5,346,421 was obtained for that design. Field tests were conducted and a few systems installed and the concept appeared promising. Unfortunately, three problems surfaced using this approach.

1. The first was economic-The circuit and latching solenoid assembly was too expensive to compete in the market place, particularly with the low production volumes involved.
2. The second problem was lack of sales representation, distribution, and marketing expertise.
3. The third was a technical problem uncovered during one of the field installations. In cases where field wires were tightly bundled over long distances, capacitive coupling could take place in an AC powered system if the current load were so low that it could

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not suppress the existing floating voltages. By the nature of the low current design, this could occur with the latching solenoid SWELL. With a limited amount of financing and marketing capability, I applied for assistance from the Department Of Energy in May of 1995 in order to re-evaluate and redesign this energy saving product and learn new marketing skills to successfully commercialize this product.

The following is a chronology of how the DOE ERIP grant assisted in the evolution from the SWELL to the BCS (Battery Control System):

1. Since the use of 24 VAC power was not practical with extremely low power loads, DC power was considered since DC has no capacitive coupling, hence no floating voltages.
2. The problem with using steady DC current is that the buried copper wire in the field reacts much like the plates of a battery, which deteriorate with use. In an irrigation system with buried wires, current flow causes galvanic action with the minerals in the ground which causes the copper wire to disintegrate.
3. At about that time, we found out that the Australian patent had been abandoned, so we could return to the original design using AC to operate a non-latching solenoid. Some effort was made toward that end, but technology demanded a better solution for broader based market acceptance.
4. Use of DC eliminates floating voltages, but not the galvanic problem. Short pulses of DC current would eliminate this galvanic effect, but all battery operated controllers that operate latching solenoids have the same limitation: The current draw of a latching solenoid is in the order of 10 to 20 times greater than that of a standard solenoid. Overall, energy is saved because of the pulsing, but the distance that a valve can be operated from the controller is severely limited to a few hundred feet, which is not sufficient distance for turf, golf, and commercial applications.
5. Therefore the solution was to use DC pulsed current in such a way that a very low current is required to activate a latching solenoid. DC pulsing for a few seconds solves both the floating voltage problem and the galvanic copper wire effect.
6. With this approach, we needed to redesign the SWELL to accommodate a pulsed DC, while simplifying the circuit to reduce cost and improve reliability. This task was accomplished with a circuit that is currently patent pending (paid for by private funds) and consists of a total of 4 passive components including the latching solenoid. This reduced cost and should improve product reliability. DOE funds were used to design, develop, and test this concept. Test results showed that the current draw is about 30 milliamps for a total of 3 seconds as compared to 250 milliamps for 30 minutes in a typical AC powered system. For all practical purposes, this is nearly a 100% energy saving compared to the original SWELL design of 70%.
7. To make the system functional and marketable, we needed a highly energy efficient battery operated controller. It so happened that simultaneously as the BCS solenoid module was being redesigned, we at Alex-Tronix had designed a single station battery operated controller with battery life in the order of 5 times greater than any on the market. This development was funded privately and is also patent pending.
8. With grant funds, we took the basic design of the single station controller, made it multiple stations with full commercial controller features, and adapted its output

circuit to operate the BCS latching circuit. This controller design is also patent pending, again financed with private funds. We called this controller-solenoid combination the BCS (Battery Control System)

To review, the grant assisted in redesigning and testing the solenoid module and with the adaptation of the battery controller to become the BCS. The results of this evolution are as follows:

1. The solenoid energy conservation has gone from 70% to virtually 100%.
2. In addition, the controller does not use any external power since it is powered with lithium batteries, so that energy saved is also nearly 100%, which is more than the energy saved by the solenoid since the controller electronics operates 24 hours a day.
3. By using 9 volt lithium batteries, the BCS can be installed anywhere in the world. Lithium batteries have a wide temperature operating range and very little self discharge characteristics. In addition, lithiums have twice the capacity of alkalines.
4. The BCS was designed in such a way that it is not limited to irrigation applications. By replacing the latching solenoid with a latching relay, lights, pumps, motors, fans, or any other form of industrial controls are now feasible, further extending the realm of energy savings.
5. Battery operation eliminates the danger of electrical fire and electrocution.
6. The BCS is installed where convenient, not where power is available.
7. Pulsed DC operation eliminates a common cause of solenoid failure due to overheated coils.
8. Battery operation virtually eliminates a common source of controller failure due to primary and secondary power and lightning surges.
9. The BCS decreases the cost of copper wire more than previously estimated.
10. Besides saving energy, low current allows operation of valves at distances up to 100 times greater than existing battery operated controllers.
11. The BCS is the only purely battery operated controller on the market. All others use solar panels in combination with batteries which save energy, but solar panels add cost, invite vandalism, are prone to malfunction and require sunlight. In addition, rechargeable batteries require monitoring and periodic replacement.
12. The BCS controller electronics battery life is 10 years, while the valve battery life should exceed 5 years or 10,000 valve actuations. This efficiency of battery life is the subject of the controller patent claims.

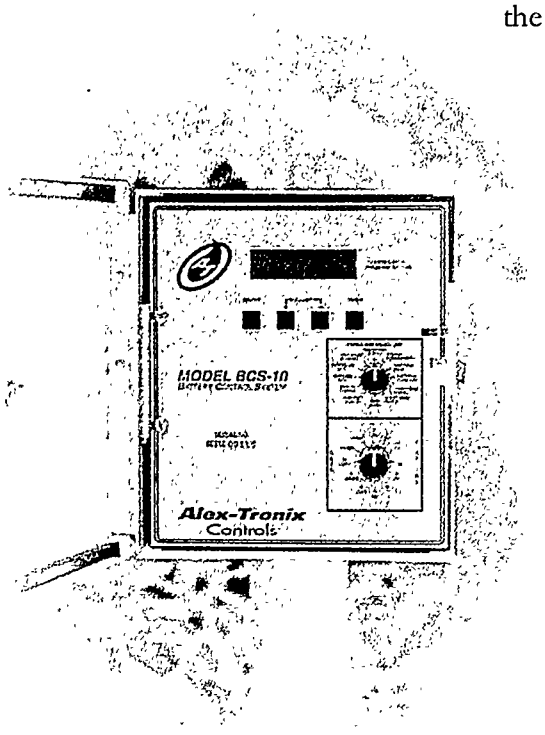
Currently, the BCS hardware is completed and the software is in the final stages of development. Three patents (funded privately) are pending. One on the controller, one on the BCS module design, and one for the BCS system. The name BCS has been submitted for a trademark. Literature is at the printer and a facsimile is included with this report. The controller plastic front panel mold and the literature were funded privately. The grant funds were primarily used for design, development, prototyping, and the marketing of the BCS. Some funds were used for facilities improvement and test equipment purchases.

Within the next 6 weeks, field tests are scheduled with production to begin in 60-90 days.

BCS (Battery Control Systems)™
A powerful, simple solution for wired control systems.

Until now, battery operated controllers had limited battery life, distance, and reliability. Everything changes with the BCS.

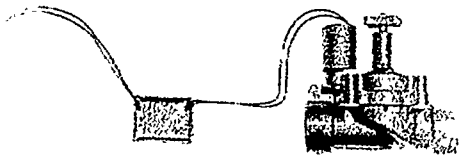
Designed under the auspices of the U.S. Department of Energy, the BCS provides longer battery life than any other battery powered controller and the ability to operate valves up to five miles away.



The BCS uses technology similar to the telegraph to send extremely low current DC pulses to valves that can be miles away. At the valve the patent pending BCS module (not a decoder) opens and closes the valve with a single positive or negative pulse from the controller. Besides incorporating proven telegraph and telephone principles, battery operation virtually eliminates electrical safety concerns -- in addition to lightning-induced controller failures common to conventional controllers. *That's reliability!*

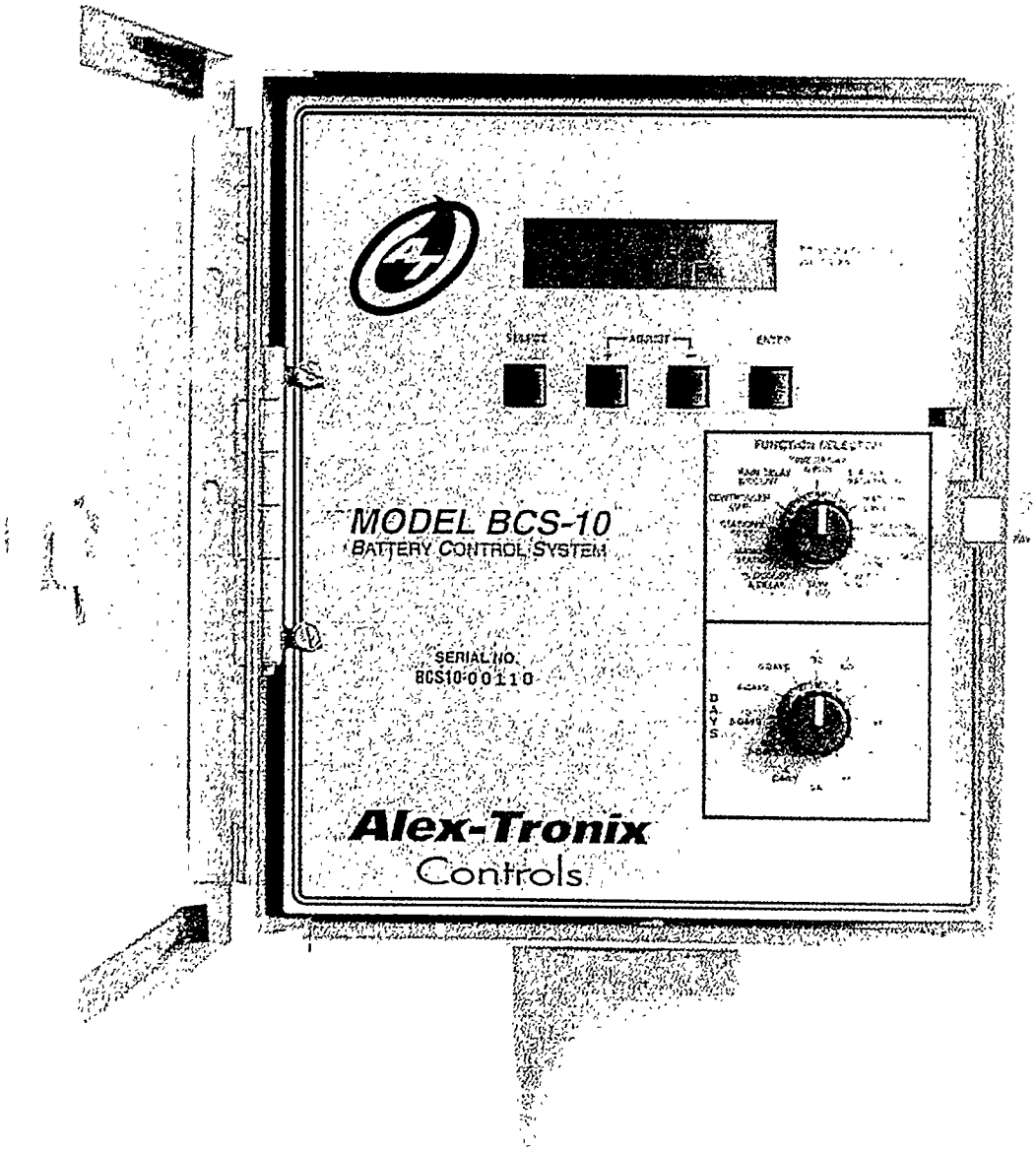
The BCS uses function selector programming. *That's simplicity!*

The patent pending BCS controllers are so energy efficient that the supplied lithium batteries power the electronics for 10 years and the valves for 5 years. *That's power!*



Patent Pending

With BCS reliability, simplicity, and power, you can set it and forget it anywhere in this new world of control!



Patent Pending

The next generation of power

BCS™ Specifications

The controller shall be a strictly battery operated multi-station unit (batteries included) with a minimum of five years of battery life. It shall be capable of operating a valve 5 miles away using 14 gauge wire. The control system shall be recognized by the U.S. Department of Energy for its energy efficiency.

BCS™ Features and Advantages

1. Five and ten station models (plus pump or master valve).
2. Ten year electronics and five year valve battery life (batteries included).
3. Five mile valve operating distance with 14 gauge (1.6 mm) wire
4. Functions and stations rotary switch programming.
5. Self powered removable panel for programming at a convenient location.
6. Two independent programs, 4 starts per day per program.
7. Automatic, semi-automatic, timed manual, and manual operation.
8. Water budget function in 10% increments from 10-200%.
9. Valve test function (invented by Alex-Tronix in 1977).
10. Watering based on days of the week or interval of days.
11. Rain delay function up to 99 days.
12. Watering up to 23 hours and 59 minutes in one minute increments.
13. Adjustable delay between stations.
14. Powder coated outdoor metal case with lock. Pedestal mounting capability.
15. Long life battery operation allows installation anywhere in the world.
16. Swing out front panel with quick connect output terminal for easy servicing with ample wiring space.
17. Easy to read high temperature rated liquid crystal display.
18. Battery operation and open output relay contacts virtually eliminate surge problems.
19. Power failure proof and electrically safe installation and troubleshooting.
20. Pulsed power eliminates burned out coils due to excessive voltage.
21. Optional solenoid module compatible with virtually any 2 wire latching solenoid.
22. Optional latching relay allows operation of lights, pumps, or other industrial controls.
23. Pulsed DC current eliminates capacitive coupling problems associated with AC powered systems and galvanic copper wire deterioration of steady DC operation.
24. Extremely low power allows use of small gauge multi-conductor cable.
25. Battery life and valve distance operation allows applications beyond non-power uses.
26. Recognized and supported by the U.S. Department of Energy for energy efficiency.
27. Patented and patents pending design.
28. Competitively priced.
29. Two year warranty.

Electrical Specifications

Electronics current in idle state: 30 microamps
Electronics current while programming: 3 milliamps
Valve current draw: 40 milliamps per valve for 1-3 seconds, 2 valves maximum
Electronics power: C size lithium battery, 7200 milliamphour
Valve power: Four 9 volt lithium batteries in series, 1200 milliamphour

Mechanical Specifications

Case size: 8.25 x 7.5 x 5.2 inches, 21.0 x 19.0 x 13.2 cm
Weight: 8.0 lb, 3.6 Kg



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A brief field test was recently conducted with the assistance of the Center for Irrigation Technology at Fresno State University.

In terms of marketing, we have established sales representatives over most of the U.S. and one in Europe. Briefcase housed demonstration units are being shipped to the sales reps with which they can demonstrate the power and abilities of the BCS. Initial designer and distributor responses have been excellent and there has been OEM (Original Equipment Manufacturer) interest. Since the BCS is not limited to DC powered applications, nor limited to the irrigation market, nor limited by electrical considerations of foreign countries, it has significant potential.

I wish to thank the Department of Energy for its financial support, project coordinators Mr Hart and Mr Jain, and the volunteer instructors of the Commercialization and Planning Workshop for their guidance and encouragement during this product evolution.

Sincerely

George Alexanian
Owner, Alex-Tronix

Set it & forget it

"Click!"



"Zzzzz."

A new world of control

Alex-Tronix Controls